Lecture Slides: Respiratory*Pathology*Test

SLIDE 1: Respiratory Pathologies and Control

• **Brain-stem respiratory centers:** Automatically adjust ventilation based on arterial CO2, O2, and pH levels.

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| **Asthma** | Bronchial smooth-muscle constriction + mucus -> ↑ airway resistance, ↓ airflow. |

| **Emphysema** | Destruction of alveolar walls reducing surface area for gas diffusion. |

• Inhalation begins when the diaphragm contracts and flattens, expanding thoracic volume.